CNO-EVIL: Beginner's Guide

1. What is CNO-EVIL?

In vanilla Nuclear Option, once something is "known", every unit of your faction sees it at its true position, every RWR points exactly at every radar, and missiles fly straight to the real target.

CNO-EVIL replaces this with a physical model. **Your faction knows only what its sensors actually find.**

- Radars follow the radar equation: power, range, the radar cross section (RCS) of the target, noise, Earth curvature and terrain.
- RCS depends on the angle you see an aircraft from. External stores and open bay doors make it larger.
- A radar that looks down at the ground can lose a target that flies across its beam (the Doppler notch).
- Jammers really fight the radar echo. Up close, the radar burns through, and a missile can home on your jammer.
- Your RWR gives a direction with an error, not a perfect bearing. How good it is depends on your aircraft.
- Everything your faction knows is a **track** with a quality **Q**. Weapons fly to the position in the track, not to the true target. A poor track means a wide miss.
- The AI plays by the same rules. It fires only when its track is good enough.

You can press **F10** at any time to switch between CNO-EVIL and vanilla and compare.

2. Why play with it

- **Hiding matters.** Terrain, the horizon, flying low, beaming a radar and the right use of a jammer really keep you alive.
- **Finding the enemy is a job.** An RWR strobe tells you only a direction. You must locate a radar before you can attack it.
- **Honest AI.** Enemy fighters and air defense react to what they detect, not to what the game knows.
- **Teamwork.** Allied radars, aircraft, ships and ground units share contacts over the datalink and warn you of missiles your own RWR cannot hear.
- **Better cockpit.** A multi-page cockpit display (PortalMFD), threat cues on the HUD, a "how well does the enemy see me" line, an IR thermal window on the helmet.
- **Modded aircraft work.** Physics needs no aircraft list, and the cockpit display is ready for modded aircraft.
- **Multiplayer.** The host runs all the physics, so every player sees the same result.

3. Core ideas in plain words

Your RWR tier

Every aircraft gets an RWR tier from its own game data. It decides how much your RWR can tell you.

	Tier 1	**Tier 2**	**Tier 3**
Examples	CI-22, A-19, SAH-46, UH-90	FS-12, UFO	FS-20, KR-67, EW-25, SFB-81
Bearing error	up to 18 deg	up to 5 deg	up to 1 deg
Hears an active radar missile at	3.5 km	10 km	10 km
Notch cue on the HUD	none	`[ BEAMING ]`	`[ NOTCH OK ]`, `[ ECM NOTCH ]`, `[ NO CLUTTER ]`
Datalink	basic	basic	advanced, shares bearings

In a Tier 1 aircraft, do not count on your RWR for missile warnings. Stay near allied radars.

Tracks and quality Q

HUD markers keep their vanilla look and behaviour, but their color shows how good the track is:

- **pale white** - rough position, most missiles will miss;
- **amber** - good enough for weapons with their own seeker;
- **red** - good for all weapons.

When contact is lost, the faction remembers the last position (aircraft 60 s, ground and naval units 180 s), and the error grows with time.

Finding an enemy radar

A strobe alone gives only a direction, and you cannot select that radar as a target yet. To locate it:

- turn about 90 degrees and fly for a few seconds, so that a second bearing crosses the first (a **fix**); or
- let an ally do it: an allied radar contact, an allied unit that sees it, or a second Tier 3 aircraft with a crossing bearing.

When the radar is located, its zone on the map turns into an icon.

Staying alive

- **Notch:** put the radar at your 3 or 9 o'clock and keep the ground behind you. Above the radar, with the sky behind you, the notch does not work. Helicopters cannot notch.
- **Jammer:** good against search radars at long range. Up close, the radar burns through. If you jam while a missile is coming and you are not in the notch, it can home on your jammer.
- **Terrain:** hills and the horizon block radar. Low and behind a ridge is the safest place.
- **Active radar missiles** keep their radar off until about 10 km from you. You get no RWR warning before that, unless an allied radar sees the missile.

4. Your first flight

1. Pick an aircraft and check its tier (section 3).
2. Look at the map (M) and the RWR page: yellow strobe = search, orange = track, red = missile guidance.
3. Watch the HUD cue on the compass and the `ID` line under the altitude.
4. Before you attack a radar, locate it with a fix. Fire when the marker is amber or red.
5. When a missile comes: notch, turn hard, descend.

HUD cue	What to do
`[ LOCK ]`	A radar tracks you. Get ready to notch.
`[ MISSILE ]`	A radar guides a missile on you. Notch that radar and descend.
`[ PITBULL ]`	An active radar missile has you. Notch now, turn hard, descend.
`[ BURN-THROUGH ]`	The radar sees through your jammer. Jammer off, notch or hide behind terrain.
`[ HOJ RISK ]`	The missile can home on your jammer. Get into the notch now.
`[ NO CLUTTER ]`	You are beaming, but the sky is behind you. Descend.
`[ DL: MISSILE <clock> ]`	An ally sees a missile at that clock position. Look and defend.

5. Cockpit displays at a glance

- **ID line** (under the altitude) - how well the enemy sees you: `ID NO` (not at all), `ID MEM` (lost you, remembers), `ID LOW` (knows something is there), `ID MED` (knows your class), `ID HIGH` (knows your type or has you on radar), `ID OPT` (sees you with eyes or optics).

- **RCS line** - your own frontal RCS. Watch it grow with external stores.
- **SNR bar** - how close the strongest enemy radar is to detecting you. Keep the pip left of the center tick.
- **Map** - RWR strobes. A cone ending in a circle means the radar is somewhere inside that zone, not at its center. A narrow cone to an icon means it is located.
- **PortalMFD** (the default TacScreen) - two pages side by side: TSD (tactical map), RADAR, RWR, EOTS (target camera), SMS (weapons), NAV (level map), HSI (flight data). EOTS opens by itself when the camera comes on. Around your aircraft on TSD, the **EM scan ring** shows received signals: a bump points to a radar, a dip inward is a radar that illuminates you but cannot detect you yet.
- **HMD thermal window** - a black-and-white IR view through the airframe in front of your head. Engines and flares show as white spots.

6. Hotkeys

Key	Action
F10	CNO-EVIL on / off (switch to vanilla to compare)
;	Next page on the left screen. With Shift: next range of that page
'	Next page on the right screen. With Shift: next range
/	Next page on the centre screen (only cockpits that have one)
.	Next page on the lower centre screen (only cockpits that have one)
] / [	Zoom the cockpit minimap in / out
B	Open / close the HMD thermal window (cockpit view, aircraft with a target camera)
V	Thermal window zoom: 1x, 5x, 10x
F9	Telemetry panel (debug)
F8	Datalink overlay (debug)
F11	Radar rays (debug)

Range steps work on TSD, NAV and RADAR. The page keys do nothing while you type in chat.

F9, F8 and F11 show data a real pilot would not have. They are for testing. In multiplayer, F8 to F11 work only in single player and on the host; a client follows the host's F10.

To change a key, close the game, open the settings file, find the key name (sections `01. General`, `02. Debug Overlays`, `13. Tactical Map & Minimap`, `14. Cockpit TacScreen`, `16. HMD Thermal Window`) and write a Unity key name, for example `F7` or `Keypad5`.

7. Modded aircraft

The physics uses no aircraft list. Radar, RWR tier, RCS and weapon RCS come from each aircraft's own game data, so an aircraft from another mod gets the full CNO-EVIL treatment automatically.

The cockpit display is the one part that needs a layout for each cockpit model, because every cockpit splits its screen texture in its own way. Layouts are ready for:

Aircraft	RWR tier	PortalMFD screens
OA-27 Cavalier	1	one main screen (two halves)
F-16M King Viper	2	three MFDs: left, right, centre
F-22E Strike Raptor	3	four MFDs: left, right, centre, lower centre
F-99 Shrike	2	three MFDs: left, right, centre
FS-41 Eclipse	3	two MFDs (tandem cockpit)
MC-260 Chimera	2	the two MFDs of the pilot seat
RAH-72 Knockout	1	one main screen (two halves)
FQ-106 Kestrel	3	one main screen (two halves)
FS-3 Ternion	2	two MFDs: left, right
KR-33 Agni	2	one main screen (two halves)

On cockpits with separate MFDs, each MFD shows its own page: `;` left, `'` right, `/` centre, `.` lower centre. Small screens of the cockpit (gauges, status strips) keep the original panels of the aircraft's author.